Fundamentals of Vision Hardware

- Resolution & FPS
  - Resolution dictates spatial detail (pixel grid count)
  - FPS dictates how many snapshots are taken per second
- Why do videos blur?
  - Motion blur happens when an object moves across the camera's sensor while the shutter is open.
  - If the shutter speed is too slow relative to the speed of the saber, the object moves across multiple pixels within a single exposure, making a "streak".

Rolling vs Global shutter

- Rolling Shutter
  - Reads the light sensors on the camera line-by line (top to bottom)
  - Causes geometric distortion (skewing) on fast moving objects, because the bottom of the frame is captured slightly later than the top.
- Global Shutter
  - Exposes every single pixel at the same time
  - Captures fast movements perfectly without warping
- Why Global Shutter Cameras are Rarer & Lower Resolution
  - Circuitry Complexity: Each pixel requires its own memory capacitor to hold data
  - High Amounts of Data: With higher resolutions, there would be too much data being outputed per second, which is why they tend to be low resolution (0.3 - 2 Mega Pixels)

Design Decisions & Justifications

Decision 1: Rolling vs Global Shutter

- Decision: Global shutter
- Justification: A saber moved by a player undergoes rapid acceleration. A rolling shutter would bend or distort the shape of the saber in the video, causing issues when estimating its position.

Decision 2: Minimum Resolution & Frame Rate

- Decision: 480p at 60 FPS
- Justification: Using high resolutions lead to extremely large volumes of data being transferred, leading to strains in processing